Task-15 : Nginx – Enabling HTTPS & Securing with Basic SSL/TLS Configuration

Goal

Enable HTTPS and secure your Nginx server by configuring SSL/TLS, HTTP → HTTPS redirection, and basic security headers.

Task Instructions

Part 1: Nginx Installation & Setup

1. Update your package list and install Nginx:

```
sudo apt update
sudo apt install nginx -y
```

2. Verify Nginx is running:

```
systemctl status nginx
```

3. Open your server's public IP in the browser — you should see the default Nginx welcome page.

Part 2: Generate a Self-Signed SSL Certificate

1. Create a directory to store SSL certificates:

```
sudo mkdir -p /etc/nginx/ssl
```

2. Generate a self-signed certificate and private key (valid for 365 days):

```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout /etc/nginx/ssl/nginx-selfsigned.key \
  -out /etc/nginx/ssl/nginx-selfsigned.crt
```

   - When prompted, you can fill out info (Country, State, etc.) or press Enter for defaults.

3. Set correct file permissions:

```
sudo chmod 600 /etc/nginx/ssl/nginx-selfsigned.key
```

Part 3: Configure Nginx for HTTPS

1. Edit the default Nginx server block:

```
sudo nano /etc/nginx/sites-available/default
```

2. Replace its contents with:

```
server {
    listen 80;
    server_name _;

    # Redirect all HTTP traffic to HTTPS
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl;
    server_name _;

    ssl_certificate /etc/nginx/ssl/nginx-selfsigned.crt;
    ssl_certificate_key /etc/nginx/ssl/nginx-selfsigned.key;

    # Basic security headers
    add_header X-Frame-Options "SAMEORIGIN";
    add_header X-Content-Type-Options "nosniff";

    root /var/www/html;
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
```

3. Test the Nginx configuration:

```
sudo nginx -t
```

4. Reload Nginx:

```
sudo systemctl reload nginx
```

Part 4: Firewall & Testing

1. Allow HTTPS through the firewall:

```
sudo ufw allow 'Nginx Full'
```

2. Test in your browser:
   - Access your server using `https://<your-server-ip>`
   - It should show the default Nginx page, but with a **self-signed SSL warning** (expected).
3. Check HTTP redirection:
   - Open `http://<your-server-ip>` → should redirect to `https://<your-server-ip>`.
4. Verify security headers:

```
curl -I https://<your-server-ip>
```

   You should see:

```
X-Frame-Options: SAMEORIGIN
```

```
X-Content-Type-Options: nosniff
```

Expected Deliverables

- Nginx server running with HTTPS enabled.
- Automatic HTTP → HTTPS redirection.
- Security headers (`X-Frame-Options`, `X-Content-Type-Options`) present in response.
- Screenshots showing:
  - HTTPS access (browser warning due to self-signed cert).
  - HTTP redirect to HTTPS.
  - `curl -I` output with security headers.

Time to Complete

2 hours